MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL,
COUNTERS, CASCADABLE, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/315C, dated 17 January 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.2.3, case outline E: Delete "14-lead" and substitute "16-lead".

* At the bottom of the page, add the following distribution statement:

"DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited."

PAGE 2

* 1.3, thermal resistance, delete and substitute: (alter change made by previous amendment)

"Thermal resistance, junction-to-case (θ_{JC}):
Cases A, B, C, D, E, F, and 2 - - - - - - - - - - - (see MIL-M-38510, appendix C)

* 1.3, junction temperature, delete and substitute: (alter change made by previous amendment)

"Junction temperature (T_J) 3/ - - - - - - - - - - - +175°C"

1.4, supply voltage: Add "(V_{CC})".

* 1.4, width of clock pulse, $t_{w(clock)}$, delete and substitute as follows:

"Width of clock pulse, $t_{w(clock)}$
Types 03, 06, 09, 11, 12, 13 - - - - - - - - - - 25 ns minimum
Type 04 - 30 ns minimum
Type 05 - 20 ns minimum"

* 1.4, setup time, $t_{(setup)}$, delete and substitute as follows:

"Setup time, $t_{(setup)}$
Types 03, 11, 12
Enable P - - - - - - - - - - - - - - - - - - 25 ns minimum
Load - 25 ns minimum
Clear (types 11 and 12 only) - - - - - - - 20 ns minimum
Type 04
Enable P - - - - - - - - - - - - - - - - - - 35 ns minimum
Load - 35 ns minimum
Data inputs
Types 03, 09 11, 12, 13 - - - - - - - - 20 ns minimum
Type 04 - - - - - - - - - - - - - - - - - 25 ns minimum
Types 07, 08 - - - - - - - - - - - - - - - 30 ns minimum

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
Check the source to verify that this is the current version before use.

Type 05	
Data, L inputs	15 ns minimum
U/D input	30 ns minimum
EP, ET inputs	15 ns minimum
Type 06	
Data, L inputs	25 ns minimum
U/D input	30 ns minimum
EP, ET inputs	25 ns minimum"

* Footnote 3/, delete as in amendment 2 and substitute the following:

"3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in according with method 5004 of MIL-STD-883."

* Footnote 4/, delete in its entirety (in amendment 2).

PAGE 3

* 1.4, hold time at any input, $t_{(hold)}$, types 05, 06, delete and austitute:

"Types 05, 06	
Data, EP, ET inputs	5 ns minimum
L, U/D inputs	0 ns minimum"

3.2.3, delete in its entirety and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request."

PAGE 4

TABLE I, low-level input current at input B, I_{IL3}, delete and substitute as below:

"

Low-level input current at input B	I_{IL3}		
01, 10	-0.4	-3.2	mA
02	-0.4	-1.6	mA

"

PAGE 6

TABLE I, F_{MAX}, t_{PHL1}, t_{PLH1}, t_{PHL2}, t_{PLH2}, t_{PHL4}, t_{PLH4}, conditions column: After "C_L = 50 pF", add "±10%".

PAGES 7 AND 8

TABLE I, delete and substitute new table I as printed on pages 6 and 7 of this amendment.

PAGE 9

* TABLE II, additional electrical subgroups for group C periodic inspections: Delete in its entirety.

3.6, second sentence, delete in its entirety.

* 4.2a (1) After test condition D, add: "E or F". (alter change made by previous amendment)

PAGE 13

* FIGURE 1, terminal connections: Delete "Device type 01" and substitute "Device type 02". Delete "Device type 02" and substitute "Device type 01".

Source: https://assist.dla.mil -- Downloaded: 20260-21T14:08Z
Check the source to verify that this is the current version before use.

PAGE 15

* FIGURE 1, terminal connections, device types 09 and 13, pin 5: Delete "ENABLE Q" and substitute "ENABLE G".

PAGE 38

FIGURE 2, delete and substitute new figure 2 as printed on page 8 of this amendment.

PAGE 55

* FIGURE 6, note: Delete "$t_{w(clear)}$ = 20 ns" and substitute "20 ns ≤ $t_{w(clear)}$ ≤ 25 ns". Delete "t_{setup} = 20 ns" and substitute "20 ns ≤ t_{setup} ≤ 25 ns".

PAGE 79

* TABLE III, V_{OH}, test number 5, Ro(1) column: Add "3/".

PAGE 81

TABLE III, footnote 7/, I_{IL1} parameter, circuit C: Delete "-120/-360" and substitute "-.12/-.36".

PAGE 94

* TABLE III, footnote 4/, I_{IL4}, Clear 03 and Clear 04, circuit F: Delete "-150/-380" and substitute "-0/-100".

* TABLE III, footnote 4/, I_{IL4}, Clear 11 and Clear 12, circuit F: Delete "-300/-760" and substitute "-150/-450".

* TABLE III, footnote 4/, I_{IL4}, A, B, C, D, circuit F: Delete "-160/-400" and substitute "-0/-100".

* TABLE III, footnote 4/, I_{IL5}, Load and EnT, circuit F: Delete "-300/-760" and substitute "-160/-400" and "-300/-760", respectively.

* TABLE III, footnote 4/, I_{IL6}, Clock, circuit F: Delete "-120/-360" and substitute "-0/-100".

PAGE 95

TABLE III, pin designation 2 column: Delete "CP" and substitute "CK".

* TABLE III, V_{OH}, test numbers 6 through 10, Q_D, Q_C, Q_B, Q_A, and Ripple carry columns: Delete "-4 mA" and substitute "-.4 mA" (5 places).

* TABLE III, V_{IC}, test number 12, measured terminal column: Delete "CP" and substitute "CK".

* TABLE III, I_{IL13}, test number 25, measured terminal column: Delete "CP" and substitute "CK".

* TABLE III, I_{IH17}, test number 30, measured terminal column: Delete "CP" and substitute "CK".

PAGE 96

TABLE III, pin designation 2 column: Delete "CP" and substitute "CK".

* TABLE III, I_{IH18}, test number 39, measured terminal column: Delete "CP" and substitute "CK".

* TABLE III, I_{OS}, test numbers 48 through 51, U/$\overline{D}$, A, B, C, D, EP, L, and ET columns: Add "-".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
Check the source to verify that this is the current version before use.

PAGE 97

TABLE III, pin designation 2 column: Delete "CP" and substitute "CK".

* TABLE III, test number 72, Ripple carry column: Delete "„" and substitute "L".

* TABLE III, test number 77, delete in its entirety.

* TABLE III, test number 81, Ripple carry column: Delete "„" and substitute "L".

* TABLE III, test number 82, Ripple carry column: Delete "„" and substitute "H".

* TABLE III, test number 88, Q_B and Q_A columns: Delete "L" and "H" and substitute "H" and "L", respectively.

* TABLE III, test number 89, Q_B and Q_A columns: Delete "„" and substitute "L" and "H", respectively.

* TABLE III, test number 93, Q_C column: Delete "„" and substitute "L".

PAGES 98 THROUGH 101

TABLE III, pin designation 2 column: Delete "CP" and substitute "CK".

PAGES 102 AND 103

* TABLE III, delete and substitute new table III as printed on pages 9 and 10 of this amendment (alter change made by previous amendment).

PAGE 104

* TABLE III, footnote 3/, I_{IL12}, circuit E: Delete "-3.0/-400" and substitute "-0.5/-400".

TABLE III, footnote 3/, I_{IL15}, circuit C: Delete "-290/-720" and substitute "-140/-720".

PAGE 105

TABLE III, V_{OL}, test numbers 5 and 6, pin 16: Add "4.5 V".

PAGE 113

TABLE III, footnote 3/, circuit B: Delete "(08 only)".

TABLE III, footnote 4/: Delete "B" and substitute "B = -15/-110".

* TABLE III, footnote 10/, delete in its entirety and substitute: "10/ Apply momentary GND, then 4.5 V minimum prior to input pulses. Maintain 4.5 V minimum for measurement."

PAGE 122

TABLE III, footnote 4/: Delete "B" and substitute "B = -15/-110".

PAGE 123

* TABLE III, I_{IH5}, minimum limit column: Add "13/".

PAGE 124

TABLE III, subgroup 9, t_{PHL2}, test number 70, maximum limit column: Delete "40" and substitute "50".

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:08Z
Check the source to verify that this is the current version before use.

PAGE 125

* TABLE III, add footnote 13/ as follows:

"13/ The minimum limit for circuit F shall be -150 μA."

PAGE 126

4.4.3b, delete in its entirety.

* 4.4.3c (1) After test condition D: Add "E or F". (alter change made by previous amendment)

PAGE 127

6.1, delete in its entirety.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1286)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Propagation delay time, low to high, clock to Q	t_{PLH5}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 2 kΩ	03, 04, 11, 12	3	41	ns
Propagation delay time, high to low, clock to Q	t_{PHL5}		$\overline{03}$, $\overline{04}$, 11, 12	3	45	ns
Propagation delay time, low to high, clock to Q	t_{PLH5}		05, 06	3	26	ns
Propagation delay time, high to low, clock to Q	t_{PHL5}		05	3	26	ns
			06	3	36	
Propagation delay time, low to high, clock (data) to Q	t_{PLH6}		03, 04, 11, 12	3	42	ns
Propagation delay time, high to low, clock (data) to Q	t_{PHL6}		03, 04, 11, 12	3	48	ns
Propagation delay time, low to high, EnT to carry	t_{PLH7}		03, 04, 11, 12	3	28	ns
Propagation delay time, high to low, EnT to carry	t_{PHL7}		03, 04 11, 12	3	28	ns
Propagation delay time, low to high, ET to RC	t_{PLH7}		05	3	18	ns
			06	3	28	
Propagation delay time, high to low, ET to RC	t_{PHL7}		05	3	28	ns
			06	3	32	
Propagation delay time, high to low, clear to Q	t_{PHL8}		03, 04, 11, 12	3	46	ns
Propagation delay time, low to high, load to Q	t_{PLH8}		07, 08	3	63	ns
Propagation delay time, high to low, load to Q	t_{PHL10}		07, 08	3	63	ns
Propagation delay time, low to high, counts up and down to Q, U/D to RC	t_{PLH9}		07, 08	3	60	ns
			05	3	26	
			06	3	32	

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T14:08Z
Check the source to verify that this is the currentversion before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Propagation delay time, high to low, counts up and down to Q, U/D to RC	t_{PHL11}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 2 kΩ	07, 08	3	73	ns
			05	3	33	
			06	3	37	
Propagation delay time, high to low, clear to Q	t_{PHL12}		07, 08	3	56	ns
Propagation delay time, low to high, load to Q	t_{PLH10}		09, 13	3	53	ns
Propagation delay time, high to low, load to Q	t_{PHL13}		09, 13	3	77	ns
Propagation delay time, low to high, clock to Q	t_{PLH11}		09, 13	3	41	ns
Propagation delay time, high to low, clock to Q	t_{PHL14}		09, 13,	3	57	ns
Propagation delay time, low to high, clock to Max / Min	t_{PLH12}		09, 13	3	66	ns
Propagation delay time, low to high, clock to ripple carry	t_{PLH12}		05	3	35	ns
			06	3	38	
Propagation delay time, high to low, clock to Max / Min	t_{PHL15}		09, 13	3	80	ns
Propagation delay time, high to low, clock to ripple carry	t_{PHL15}		05	3	37	ns
			06	3	40	

1/ Use $I_{OL} + I_{IL3(MAX)}$ for V_{OL} test on Q_A.

2/ Not more than one output should be shorted at a time.

3/ I_{CCH} is measured: (a) With the load input high; and (b) Then again with the load input low with all other inputs high and all outputs open.

4/ I_{CCL} is measured: (a) With the clock input high; and (b) Then again with the clock input low with all other inputs low and all outputs open.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
Check the source to verify that this is the current version before use.

Device type 12

Circuit B

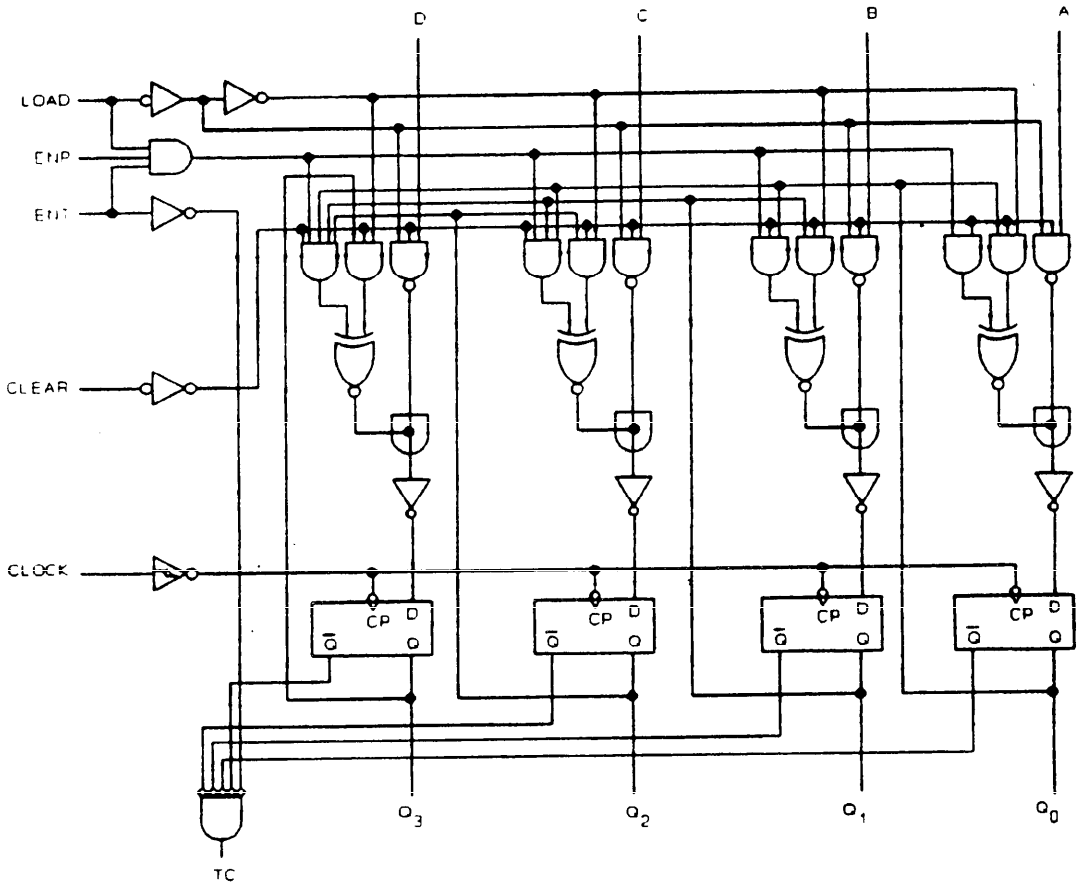


FIGURE 2. Logic diagram - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12.	13	14	15	17	18	19	20		Min	Max	
			Test no.	U/$\overline{D}$	CK	A	B	C	D	EP	GND	L	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
9 T_C = +25°C	t_{PLH5}	See fig. 7	129	5.0 V	IN 7/	GND	GND	GND	GND	GND	GND	IN	GND						5.0 V	CK to Q_A	3	22	ns
		"	130	"	"	5.0 V	"	"	"	"	"	"	"			OUT	OUT		"	CK to Q_B	"	"	"
		"	131	"	"	"	5.0 V	"	"	"	"	"	"		OUT				"	CK to Q_C	"	"	"
		"	132	"	"	"	"	5.0 V	"	"	"	"	"	OUT					"	CK to Q_D	"	"	"
	t_{PHL5}	"	133	"	" 7/	5.0 V	GND	GND	GND	"	GND	IN	GND						"	CK to Q_A	"	32	"
		"	134	"	"	"	5.0 V	"	"	"	"	"	"			OUT	OUT		"	CK to Q_B	"	"	"
		"	135	"	"	"	"	5.0 V	"	"	"	"	"		OUT				"	CK to Q_C	"	"	"
		"	136	"	"	"	"	"	5.0 V	"	"	"	"	OUT					"	CK to Q_D	"	"	"
	t_{PHL15}	"	137	"	"	GND	"	"	"	"	"	"	"					OUT	"	CK to RC	"	35	"
	t_{PLH12}	"	138	"	"	GND	"	"	"	"	"	5.0 V	"					OUT	"	CK to RC	"	33	"
	t_{PLH5}	"	139	GND	"	5.0 V	GND	GND	GND	"	"	5.0 V	"				OUT		"	CK to Q_A	"	22	"
	t_{PHL5}	"	140	"	"	GND	GND	"	"	"	"	"	"				OUT		"	CK to Q_A	"	32	"
	t_{PLH5}	"	141	"	"	"	5.0 V	"	"	"	"	"	"			OUT			"	CK to Q_B	"	22	"
	t_{PHL5}	"	142	"	"	"	GND	"	"	"	"	"	"			OUT			"	CK to Q_B	"	32	"
	t_{PLH5}	"	143	"	"	"	"	5.0 V	"	"	"	"	"		OUT				"	CK to Q_C	"	22	"
	t_{PHL5}	"	144	"	"	"	"	GND	"	"	"	"	"		OUT				"	CK to Q_C	"	32	"
	t_{PLH5}	"	145	"	"	"	"	"	5.0 V	"	"	"	"	OUT					"	CK to Q_D	"	22	"
	t_{PHL5}	"	146	"	"	"	"	"	GND	"	"	"	"	OUT					"	CK to Q_D	"	32	"
	t_{PHL7}	"	147	5.0 V	" 7/	5.0 V	5.0 V	5.0 V	5.0 V	"	"	5.0 V	IN					OUT	"	ET to RC	"	28	"
	t_{PLH7}	"	148	5.0 V	"	"	"	"	"	"	"	5.0 V	IN					OUT	"	ET to RC	"	24	"
	t_{PHL11}	"	149	IN	IN 7/	"	"	"	"	"	"	5.0 V	GND					OUT	"	U/$\overline{D}$ to RC	"	32	"
	t_{PLH9}	"	150	IN	IN	"	"	"	"	"	"	5.0 V	GND					"	"	U/$\overline{D}$ to RC	"	28	"
	t_{PHL7}	"	151	GND	IN 7/	GND	GND	GND	GND	"	"	5.0 V	IN					"	"	ET to RC	"	28	"
	t_{PLH7}	"	152	GND	IN	"	"	"	"	"	"	5.0 V	IN					"	"	ET to RC	"	24	"
	t_{PHL11}	"	153	IN	IN 7/	"	"	"	"	"	"	5.0 V	GND					"	"	U/$\overline{D}$ to RC	"	32	"
	t_{PLH9}	"	154	IN	IN	"	"	"	"	"	"	5.0 V	"					"	"	U/$\overline{D}$ to RC	"	28	"
	F_{MAX}	"	155	5.0 V	IN	"	"	"	"	"	"	5.0 V	"	OUT	OUT	OUT	OUT		"	CK to Q_n	25		MHz
	F_{MAX}	"	156	GND	IN	5.0 V	5.0 V	5.0 V	5.0 V	"	"	5.0 V	"	OUT	OUT	OUT	OUT		"	CK to Q_n	25		MHz

See footnotes at end of device types 05 and 06.

Source: https://assist.dla.mil -- Downloaded: 20260-21T14:08Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$U/\overline{D}$	CK	A	B	C	D	EP	GND	L	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
10 T_C = +125°C	t_{PLH5}	See fig. 7	157																	CK to Q_A	3	26	ns
		"	158																	CK to Q_B	"	"	"
		"	159																	CK to Q_C	"	"	"
		"	160																	CK to Q_D	"	"	"
	t_{PHL5}	"	161																	CK to Q_A	"	36	"
		"	162																	CK to Q_B	"	"	"
		"	163																	CK to Q_C	"	"	"
		"	164																	CK to Q_D	"	"	"
	t_{PHL15}	"	165																	CK to RC	"	40	"
	t_{PLH12}	"	166																	CK to RC	"	38	"
	t_{PLH5}	"	167																	CK to Q_A	"	26	"
	t_{PHL5}	"	168																	CK to Q_A	"	36	"
	t_{PLH5}	"	169																	CK to Q_B	"	26	"
	t_{PHL5}	"	170																	CK to Q_B	"	36	"
	t_{PLH5}	"	171																	CK to Q_C	"	26	"
	t_{PHL5}	"	172																	CK to Q_C	"	36	"
	t_{PLH5}	"	173																	CK to Q_D	"	26	"
	t_{PHL5}	"	174																	CK to Q_D	"	36	"
	t_{PHL7}	"	175																	ET to RC	"	32	"
	t_{PLH7}	"	176																	ET to RC	"	28	"
	t_{PHL11}	"	177																	$U/\overline{D}$ to RC	"	37	"
	t_{PLH9}	"	178																	$U/\overline{D}$ to RC	"	32	"
	t_{PHL7}	"	179																	ET to RC	"	32	"
	t_{PLH7}	"	180																	ET to RC	"	28	"
	t_{PHL11}	"	181																	$U/\overline{D}$ to RC	"	37	"
	t_{PLH9}	"	182																	$U/\overline{D}$ to RC	"	32	"
	F_{MAX}	"	183																	CK to Q_n	25		MHz
	F_{MAX}	"	184																	CK to Q_n	25		MHz
11	Same tests, conditions and limits as for subgroup 10 except T_C = -55°C and V_{CC} = 4.5 V for F_{MAX}.																						

Same conditions as for subgroup 9.

See footnotes at end of device types 05 and 06.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
Check the source to verify that this is the current version before use.